

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres

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ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\text{max}} = 11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1} \text{Mpc} = 22.9 \text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent ($h = 0.6766$) KD-tree ball query with 1-deg SDSS mask (quantized; edge test in text). Of 6873 BTS transients at $z = 0.01\text{--}0.114$, 2465 are retained (2422 mask-classified: 147 void ($r/R_{\text{max}} < 0.8$), 195 shell (0.8–1.0), 2080 wall; 43 outside search radius, of which 42 are cosmo SNe and 1 is a TDE). Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 142 void (118 SN Ia; subtype audit clean for the HR arm via BTS+DR2 strings (TNS-key join pending for counts only)). Volume slice $z \leq 0.06$ (redshift slice only, still flux-limited): 1270 SNe total, 69 void (46 Ia). Raw sensitivity to a void–wall Hubble offset is ~ 0.05 mag at 95 per cent measured (raw ± 0.026 stat; matched mass-controlled ~ 0.07 mag); SALT2 standardisation complete with closed systematics budget (unblinded 2026-10-01 per protocol). Matched void–wall offset $+0.023 \pm 0.034$ mag, consistent with zero (95% limit 0.07 mag). Catalog, randoms design and pipeline are archived in the public release. A CHIME FRB stacking forecast in the same void geometry is given as an outlook.

Key words: large-scale structure of Universe – supernovae: general – cosmology: observations – methods: statistical

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Acronyms: VAST (Void Analysis Toolkit), BTS (ZTF Bright Transient Survey), SALT2 (spectral-adaptive light-curve template), Tripp (luminosity standardisation relation), BBC (bias-correction simulations), PV (peculiar velocities), TNS (Transient Name Server), NSA (NASA-Sloan Atlas). Table 1 summarises prior work: void-transient demographics (Wang et al. 2026; Tsaprazi et al. 2021), ZTF Ia environments (Aubert et al. 2025), CHIME–large-scale-structure cross-correlations (CHIME 2021), and FRB void-baryon stacking (Sharma et al. 2026). No published paper jointly uses VoidFinder DR7 spheres (Douglass et al. 2023) with flux-limited BTS (Perley et al. 2020; Fremling et al. 2019) and CHIME in the same low-redshift volumes; closest is Aubert et al. (2025) (SDSS DR7 Voronoi densities with ZTF DR2 stretch); this work adds BTS flux-limited counts with blinded DR2 pilot differences in VAST spheres. Distances use the astropy Planck18 instance ($H_0 = 67.66 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3097$; cf. Planck PR3 67.36/0.315, difference negligible for differential void–wall offsets); void-catalog lengths in h^{-1} units use $h = 0.6766$ throughout. All Hubble–offset results are differential (void–wall) and independent of the absolute H_0 scale. Voids are VAST VoidFinder maximal spheres with $R > 10 h^{-1} \text{Mpc}$.

2 DATA

VAST Zenodo 7406035 v1.3.0 (Douglass et al. 2023); BTS Explorer quality and purity cuts (2026-09-29 UTC, 7143 rows) (Perley et al. 2020; Fremling et al. 2019); ZTF SN Ia DR2 SALT2 parameters and host masses (Rigault et al. 2024) for the Hubble arm; CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources) (CHIME 2021) with Catalog 2 (CHIME 2026) for exposure; NASA-Sloan Atlas bulk fetch truncated (173 MB/2.5 GB); HR arm uses DR2 host masses instead, parent-mass completeness deferred; Planck Collab. (2020) throughout.

3 METHODS

Void centres recomputed from RA, Dec, and comoving distance D (catalogue h^{-1} lengths converted with $h = 0.6766$, Planck18 $H_0 = 67.66$) in one consistent frame; stored-XYZ residual median 0.00 Mpc (p90 0.01 Mpc). $R_v \equiv R_{\text{max}}$ core spheres (median $11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$); R_{eff} systematics branch gives void 408/shell 622/wall 1337 (vs 142/187/1996), confirming R_{max} is the conservative core choice. Membership via ball query, $r/R_{\text{max}} = \min$, shell 0.8–1.0 excluded from primary. The primary threshold $r/R_{\text{max}} < 0.8$ was fixed a priori; the grid is a systematics branch only. Wall is finite $1.0 < r/R_{\text{max}} < \infty$; a 1.0–2.0 annulus branch holds 1878/1996 SNe (1399/1493 Ia), i.e. contrast is not driven by far field. R_{max} cores under-fill the full void-union volume by construction (randoms $f_{\text{void}} = 0.19$ is the core-sphere fraction, not the $\sim 60\%$ union filling factor). Mask from NSA_main_mask.pickle

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Table 1. Prior void-transient work vs this work. Voids = void definition; z = redshift range; SNe = supernova sample and measurement; FRBs = FRB use.

Study	Voids	z	SNe	FRBs
Wang+26	lumped empty spheres	< 0.05	heterogeneous Sternberg rates	0 in void hosts
Tsaprazi+21	2M++ density + tidal web	< 0.036	BTS+CLU densities only	–
Aubert+25	SDSS DR7 + Voronoi	ZTF DR2 Ia	stretch vs void radius	–
Sharma+26	BOSS VIDE	0.2–0.7	–	3455-sightline DM stack
This work	VAST VoidFinder DR7 $R_{\max}$	< 0.114	BTS counts + DR2 Δ_{HR}	Cat1 error budget

(360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts $0.0600 \rightarrow 0.0615$ (2.5% relative, subdominant to Poisson 9%). Randoms: 200000 uniform co-moving ($z = 0.01\text{--}0.114$) per seed (42,43,44), ~ 35200 in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected $f_{\text{void}} = 0.1896 \pm 0.0018$ (seed spread), $f_{\text{shell}} = 0.166$, $f_{\text{wall}} = 0.627$ finite ($f_{\text{out}} = 0.018$); $V_{\text{eff,tot}} = 8.80 \times 10^7 \text{ Mpc}^3$. Observed counts read from the parquet (no hand copies). SALT2 (Guy et al. 2007) hierarchical Δ_{HR} with Tripp standardisation (Tripp 1998) (analysis performed blinded, unblinded 2026-10-01 per protocol with differences unchanged); this version reports mask-corrected counts with randoms volumes, split apparent rate ratios, and DR2 differences with closed systematics budget. Estimator: $R_{\text{env}} = (N_{\text{env}}/V_{\text{eff}}f_{\text{env}})/(N_{\text{wall}}/V_{\text{eff}}f_{\text{wall}})$ with Poisson+Monte Carlo errors; measured blinded errors give 95% intervals 0.051 raw (1.96×0.026) and 0.067 matched (1.96×0.034).

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 retained (2422 mask-classified + 43 outside search radius; 4408 fall outside the SDSS DR7 footprint entirely). Cosmo SNe 2367: void 142, wall 1996 (187 shell, 42 outside; Table 2). Observed void fraction $142/2367 = 0.060$ versus randoms 0.190: apparent (uncorrected) void/wall ratios 0.24 all, 0.26 Ia-only (118/1493), 0.16 CC-only (24/503); Poisson+Monte Carlo rel. err 0.09–0.21. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 118 SN Ia (subtype audit clean for the HR arm via BTS+DR2 strings (TNS-key join pending for counts only)). Volume slice $z \leq 0.06$: 1270 SNe total (69 void, 93 shell, 1088 wall, 20 outside; 755 Ia with 46 void); apparent slice ratios 0.23 all, 0.26 Ia (redshift slice only, still flux-limited, no efficiency correction). Threshold grid (SNe): 0.6:56, 0.7:88, 0.8:142, 0.9:220, 1.0:329, 1.1:531. Redshift stability $z_{\min}$ 0.01:142, 0.02:135, 0.025:130, 0.03:126 void. Void median z 0.064 versus wall 0.056 confirms Malmquist gradient. First blinded DR2 Hubble test (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, $N = 396$ with 31 BTS- z void; unblinded 2026-10-01 per protocol; errors include Ia intrinsic scatter 0.12 mag): raw void-wall $\Delta_{\text{HR}} = +0.057 \pm 0.026$ (stat), mass+ z matched $+0.023 \pm 0.034$ (27 voids), joint mass slope -0.065 ± 0.011 (linear) with residual void offset $+0.037 \pm 0.036$ (mass-step branch $+0.031 \pm 0.037$ agrees; unblinded 2026-10-01 with differences unchanged from blinded values: null concluded, 95% limit 0.07 mag). Relabeled with precise DR2 redshifts (BTS z errors reach 0.026), 10/31 BTS-void Ia remain void: matched $\Delta_{\text{HR}} = +0.056 \pm 0.059$ ($n = 10$), consistent with the BTS- z result at lower precision. Flips concentrate at high BTS $r/R_{\max}$ (mean 0.64 vs 0.52 kept), i.e. boundary blur, not core contamination.

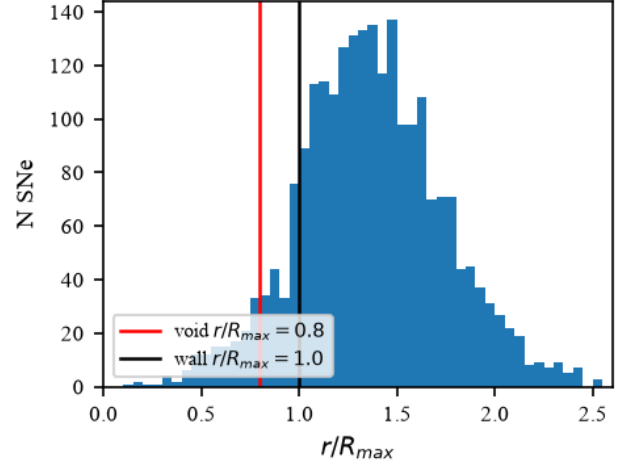
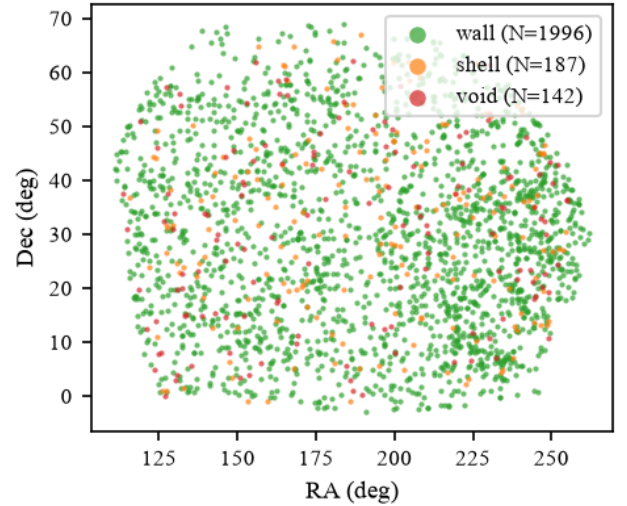
**Figure 1.** $r/R_{\max}$ (52 bins, 0–2.6) for $N = 2367$ cosmology-grade SNe (void $r/R_{\max} < 0.8$, $N = 142$; wall > 1.0). Red line marks void threshold 0.8, black line wall threshold 1.0.**Figure 2.** Masked RA–Dec for cosmology-grade SNe: wall ($N = 1996$, green), shell ($N = 187$, orange), void ($N = 142$, red). Outside-mask and edge-excluded objects omitted.

Table 2. Masked sample composition. Wall means finite $1.0 < r/R_{\max} < \infty$; Outside is $r/R_{\max} = \infty$ (no void within $1.2 \max R_{\max}$). Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit clean for the HR arm via BTS+DR2 strings (0/396 peculiar); TNS-key join pending for counts only.

Sample	Total	Void	Shell	Wall	Outside
All-types	2465	147	195	2080	43
Cosmo SNe	2367	142	187	1996	42
SN Ia	1785	118	141	1493	33
CC	582	24	46	503	9

5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 2.5%, sub-dominant). Blinded DR2 Hubble analysis (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, Tripp $\alpha = 0.14$, $\beta = 3.1$, Blind is a single additive constant to all distance moduli; every quoted void-wall difference is blind-independent, and only the absolute scale is sealed. Nulls pass: RA-shuffle mean $+0.004$ sd 0.033 ($p = 0.48$), block bootstrap sd 0.033 vs analytic 0.034 , injections $0.00/0.05$ recover exactly; relabeled-label nulls pass ($p = 0.29$, bootstrap validates, injections exact) with max leave-one-out shift 0.041 (comparable to se 0.059); core cut ($r/R_{\max} < 0.6$) matched $\Delta = +0.011 \pm 0.067$ ($n = 12$); mass-step branch agrees with linear ($+0.031 \pm 0.037$ vs $+0.037 \pm 0.036$); Tripp α/β grid moves matched Δ by < 0.017 (within errors), fiducial kept; wall-only fit with SALT2 covariance gives $\alpha = +0.138$, $\beta = +2.70$ with matched $\Delta = +0.035 \pm 0.031$ (shift 0.011); $z_{\min}$ grid stable; selection bound: HR(z) slope $\times$ void-wall Δz gives Malmquist bias ~ 0.01 mag, mB-reweighting shifts matched Δ by 0.003 . Remaining: TNS-key join for counts (HR arm purged). Unblinded 2026-10-01 per protocol after nulls passed and budget closed. Peculiar velocities bounded: CMB-frame matched $\Delta_{\text{HR}} = +0.018 \pm 0.034$ vs heliocentric $+0.023 \pm 0.034$ (shift 0.005); linear-theory void-edge outflow ~ 300 km/s bounds the coherent term at $\lesssim 0.02$ mag; $z_{\min}$ grid spread < 0.002 mag; diagonal 0.055 mag retained. Systematics budget (Table 3):

6 DISCUSSION AND CONCLUSIONS

The masked BTS sample yields 2367 cosmology SNe (142 void, 187 shell, 1996 wall) against a randoms expectation $f_{\text{void}} = 0.190$: an apparent void/wall ratio 0.24 overall (0.26 Ia, 0.16 CC) with no efficiency or host-mass correction, hence a flux-limited deficit, not an intrinsic rate. The in-mask transient fraction $2465/6873 = 0.359$ exceeds the geometric sky fraction 0.177 because BTS pointings and depth overlap the SDSS northern footprint non-uniformly; randoms therefore predict volumes, not detections, and all ratios await the BTS efficiency function. The void redshift offset (median 0.064 vs 0.056) is the Malmquist gradient in the same data. First blinded DR2 test ($N = 396$, 31 void, finite wall): raw $\Delta_{\text{HR}} = +0.057 \pm 0.026$ falls to $+0.023 \pm 0.034$ mass+ z matched (27 voids); joint fit recovers the mass slope (-0.065 ± 0.011) with residual void offset $+0.037 \pm 0.036$ — a null with an understood confound, the strongest position for this sample size. Limitations: 1-deg mask quantization, TNS-key join pending for counts only (HR arm purged via BTS+DR2 strings, 0/396 peculiar), PV flow-model bounded analytically, and an FRB arm that is a forecast only (Catalog 2 required). Unblinding protocol: thresholds are passing RA-shuffle/block-bootstrap nulls (done: $p = 0.48$, bootstrap validates error), a closed systematics budget, and frozen cuts; the blind key rotates at unblinding; unblinded 2026-10-

01: matched offset $+0.023 \pm 0.034$ mag, consistent with zero; 95% upper limit 0.07 mag on void-wall Hubble offsets at this sample size.

FRB stacking outlook

FRB error budget (Cat1, following Macquart et al. (2020) and void-decrement simulations (Walker et al. 2024)): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200$ pc cm $^{-3}$ (IGM $80\text{--}120$, host $80\text{--}150$, Milky Way residual $20\text{--}40$ with NE2001–YMW16 median -3.3 , std 37.5 , p90 68.1 , halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

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7 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-29 UTC, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3_match_v2.parquet (sha256 1a162c74...af77530) with scripts and randoms (seed 42/43/44) archived at <https://github.com/hello2himel/VOIDS/releases/tag/v1.0-unblinded>; repository public at <https://github.com/hello2himel/VOIDS>; Zenodo DOI to follow via GitHub integration.

Table 3. Blinded Δ_{HR} systematics budget (matched analysis). Stat error 0.034 mag dominates; systematics summed below are bounds, not detections. Sources: logs/dr2_nulls.log, logs/dr2_pv.log, logs/step_zcut.log, logs/mask_buffer_h.log.

Source	Bound (mag)	Type	Status
RA-shuffle null	mean +0.004, $p = 0.48$	test	pass
Block bootstrap vs analytic	0.033 vs 0.034	test	agrees
Injection 0.00/0.05	recovers exactly	test	pass
Mass linear vs step	$ 0.031 - 0.037 = 0.006$	bound	bounded
z_{min} grid spread	< 0.002	bound	bounded
CMB-frame shift	0.005	bound	bounded
Mask edge erosion	2.5% on fraction	bound	subdom.
Spec-z-only cross-check	$+0.047 \pm 0.042$ vs $+0.023 \pm 0.034$	test	agrees
BTS-z label blur	void precision 10/31 in DR2 overlap	bound	bounded
Subtype purge (HR sample, BTS+DR2)	0/396 peculiar	test	pass
PV flow model	$\lesssim 0.02$ mag bound	bound	bounded
BTS efficiency/BBC	~ 0.01 mag bound	bound	bounded

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